MSG No. 134.472 $P4_2/nnm1'$ [Type II, tetragonal]

Table 1: Wyckoff site: 2a, site symmetry: -42m1'

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[1,4,5,6,10,11,15,16,17,20,21,22,26,27,31,32]
2	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[2,3,7,8,9,12,13,14,18,19,23,24,25,28,29,30]

Table 2: Wyckoff site: 2b, site symmetry: -42m1'

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[1,4,5,6,10,11,15,16,17,20,21,22,26,27,31,32]
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[2,3,7,8,9,12,13,14,18,19,23,24,25,28,29,30]

Table 3: Wyckoff site: 4c, site symmetry: 222.1'

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[1,4,5,6,17,20,21,22]
2	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[2,3,7,8,18,19,23,24]
3	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[9,12,13,14,25,28,29,30]
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[10,11,15,16,26,27,31,32]

Table 4: Wyckoff site: 4d, site symmetry: 2.221'

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	[1,6,7,8,17,22,23,24]
2	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[2,3,4,5,18,19,20,21]
3	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	[9,14,15,16,25,30,31,32]
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[10,11,12,13,26,27,28,29]

Table 5: Wyckoff site: 4e, site symmetry: ..2/m1'

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	[1,7,9,15,17,23,25,31]
2	$\left[\frac{1}{2}, 0, 0\right]$	[2,5,10,13,18,21,26,29]
3	$\left[0, \frac{1}{2}, 0\right]$	[3,4,11,12,19,20,27,28]
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[6,8,14,16,22,24,30,32]

Table 6: Wyckoff site: 4f, site symmetry: ..2/m1'

No.	position	mapping
1	[0, 0, 0]	[1,7,9,15,17,23,25,31]
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[2,5,10,13,18,21,26,29]
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[3,4,11,12,19,20,27,28]

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[6,8,14,16,22,24,30,32]

Table 7: Wyckoff site: 4g, site symmetry: 2.mm1'

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	[1,6,15,16,17,22,31,32]
2	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	[2,3,12,13,18,19,28,29]
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	[4,5,10,11,20,21,26,27]
4	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	[7,8,9,14,23,24,25,30]

Table 8: Wyckoff site: 8h, site symmetry: 2..1'

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	[1,6,17,22]
2	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	[2,3,18,19]
3	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	[4,5,20,21]
4	$\left[\frac{1}{4}, \frac{1}{4}, -z\right]$	[7,8,23,24]
5	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	[9,14,25,30]
6	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	[10,11,26,27]
7	$\left[\frac{3}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	[12,13,28,29]
8	$\left[\frac{3}{4}, \frac{3}{4}, z\right]$	[15,16,31,32]

Table 9: Wyckoff site: 8i, site symmetry: .2.1'

No.	position	mapping
1	$\left[x, \frac{1}{4}, \frac{3}{4}\right]$	[1,4,17,20]
2	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	[2,7,18,23]
3	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	[3,8,19,24]
4	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{3}{4}\right]$	[5,6,21,22]
5	$\left[-x, \frac{3}{4}, \frac{1}{4}\right]$	[9,12,25,28]
6	$\left[\frac{3}{4}, -x, \frac{3}{4}\right]$	[10,15,26,31]
7	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	[11,16,27,32]
8	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[13,14,29,30]

Table 10: Wyckoff site: 8j, site symmetry: .2.1'

No.	position	mapping
1	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	[1,4,17,20]
2	$\left[\frac{1}{4}, x, \frac{3}{4}\right]$	[2,7,18,23]
3	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{3}{4}\right]$	[3,8,19,24]
4	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{1}{4}\right]$	[5,6,21,22]
5	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	[9,12,25,28]

continued ...

Table 10

No.	position	mapping
6	$\left[\frac{3}{4}, -x, \frac{1}{4}\right]$	[10,15,26,31]
7	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{1}{4}\right]$	[11,16,27,32]
8	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	[13,14,29,30]

Table 11: Wyckoff site: 8k, site symmetry: ..21'

No.	position	mapping
1	$[x, x, 0]$	[1,7,17,23]
2	$\left[\frac{1}{2}-x, x, \frac{1}{2}\right]$	[2,5,18,21]
3	$\left[x, \frac{1}{2}-x, \frac{1}{2}\right]$	[3,4,19,20]
4	$\left[\frac{1}{2}-x, \frac{1}{2}-x, 0\right]$	[6,8,22,24]
5	$[-x, -x, 0]$	[9,15,25,31]
6	$\left[x+\frac{1}{2}, -x, \frac{1}{2}\right]$	[10,13,26,29]
7	$\left[-x, x+\frac{1}{2}, \frac{1}{2}\right]$	[11,12,27,28]
8	$\left[x+\frac{1}{2}, x+\frac{1}{2}, 0\right]$	[14,16,30,32]

Table 12: Wyckoff site: 8l, site symmetry: ..21'

No.	position	mapping
1	$\left[x, x, \frac{1}{2}\right]$	[1,7,17,23]
2	$\left[\frac{1}{2}-x, x, 0\right]$	[2,5,18,21]
3	$\left[x, \frac{1}{2}-x, 0\right]$	[3,4,19,20]
4	$\left[\frac{1}{2}-x, \frac{1}{2}-x, \frac{1}{2}\right]$	[6,8,22,24]
5	$\left[-x, -x, \frac{1}{2}\right]$	[9,15,25,31]
6	$\left[x+\frac{1}{2}, -x, 0\right]$	[10,13,26,29]
7	$\left[-x, x+\frac{1}{2}, 0\right]$	[11,12,27,28]
8	$\left[x+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$	[14,16,30,32]

Table 13: Wyckoff site: 8m, site symmetry: ..m1'

No.	position	mapping
1	$[x, -x, z]$	[1,15,17,31]
2	$\left[x+\frac{1}{2}, x, z+\frac{1}{2}\right]$	[2,13,18,29]
3	$\left[-x, \frac{1}{2}-x, z+\frac{1}{2}\right]$	[3,12,19,28]
4	$\left[x, x+\frac{1}{2}, \frac{1}{2}-z\right]$	[4,11,20,27]
5	$\left[\frac{1}{2}-x, -x, \frac{1}{2}-z\right]$	[5,10,21,26]
6	$\left[\frac{1}{2}-x, x+\frac{1}{2}, z\right]$	[6,16,22,32]
7	$[-x, x, -z]$	[7,9,23,25]
8	$\left[x+\frac{1}{2}, \frac{1}{2}-x, -z\right]$	[8,14,24,30]

Table 14: Wyckoff site: `16n`, site symmetry: `11'`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1,17]`
2	$\left[\frac{1}{2}-y,\ x,\ z+\frac{1}{2}\right]$	`[2,18]`
3	$\left[y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[3,19]`
4	$\left[x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	`[4,20]`
5	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}-z\right]$	`[5,21]`
6	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	`[6,22]`
7	$[y,\ x,\ -z]$	`[7,23]`
8	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ -z\right]$	`[8,24]`
9	$[-x,\ -y,\ -z]$	`[9,25]`
10	$\left[y+\frac{1}{2},\ -x,\ \frac{1}{2}-z\right]$	`[10,26]`
11	$\left[-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[11,27]`
12	$\left[-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[12,28]`
13	$\left[x+\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$	`[13,29]`
14	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	`[14,30]`
15	$[-y,\ -x,\ z]$	`[15,31]`
16	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z\right]$	`[16,32]`